

Text messaging while driving increases the risk of a motor vehicle accident more than twenty-fold. The number of deadly highway accidents has spiked after decades of decline, mainly because of texting while driving. Despite this, nearly 40% of American drivers report having texted while driving and 20% of drivers report doing so regularly. Motor vehicle traffic accidents are the leading cause of death among Americans age 16 to 25.

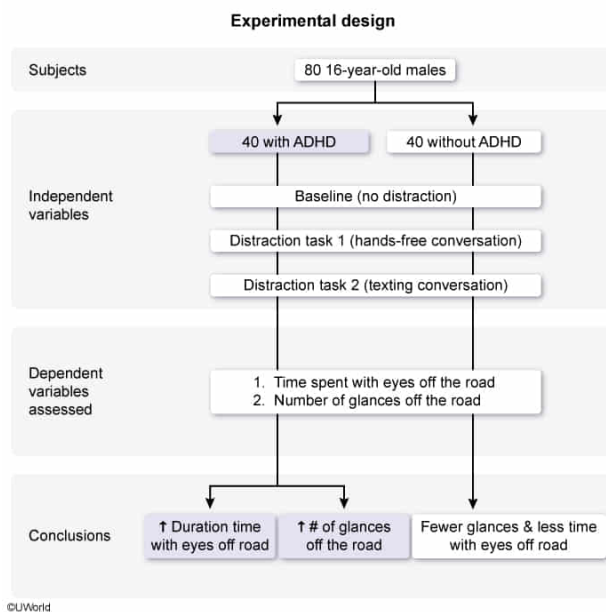
Researchers conducted a study to investigate the extent to which driving is impaired by prolonged gazes away from the highway (as a correlate to texting while driving) for males with and without attention-deficit hyperactivity disorder (ADHD). Eighty 16-year-old males with a valid driver's license were recruited for the study; half had ADHD (26 diagnosed as mild, 14 diagnosed as moderate) and half did not. Each participant was asked to complete six driving tasks (eg, avoiding an object on the road, merging, making unexpected stops) using a highly realistic driving simulator that projects computer-generated, two-dimensional, 360° visuals, requiring the use of side and rearview mirrors. The baseline condition included no distraction. In the first distraction condition, participants drove while conducting a hands-free phone conversation, and in the second distraction condition they were texting while driving. Two dependent variables were assessed: (1) percentage of driving time with eyes diverted from the road, and (2) number of glances away from the road lasting longer than two seconds.

Under both distraction conditions, participants with ADHD (compared to those without ADHD) demonstrated a significantly higher percentage of driving time with eyes diverted from the road, as well as a significantly higher number of prolonged glances away from the road. In addition, unsteady lane position while driving was correlated with both ADHD and texting. The researchers concluded that impairments in driving performance were due to the extreme cognitive workload of competing tasks and visual inattention.

Which of the following conclusions about ADHD is most supported by this study?

- ☐ A. Extreme cognitive workload increases impulsivity for those with ADHD. [8%]
- ✓ ☒ B. Distraction increases visual inattention for those with ADHD. [74%]
- ☐ C. Texting while driving has a similar effect on driving as does having ADHD. [14%]
- ☐ D. Teenagers with ADHD are worse drivers than those without ADHD. [3%]

Explanation:



Attention-deficit hyperactivity disorder (ADHD) is characterized by **inattention** and **impulsivity**. Inattentive symptoms include easy distractibility and difficulty focusing on tasks. Impulsive symptoms include behaviors that are not carefully thought out.

The independent (researcher-controlled) variables in this study included: presence or absence of ADHD and the three driving tasks (no distraction and two distraction conditions). Under both distraction conditions (hands-free calling and texting), when compared to controls, participants with ADHD diverted their eyes from the road more frequently and for longer periods of time. This suggests that *distraction increases visual inattention for those with ADHD*.

(Choice A) While it is true that people with ADHD are more likely than people without ADHD to make risky, impulsive decisions, impulsivity was not assessed in this study.

(Choice C) The purpose of the study was to investigate how much a distracting task can impair the driving of males with and without ADHD. The study design does not support the conclusion that texting while driving has a similar effect on driving as does having ADHD.

(Choice D) This study assessed how much a distracting task impaired driving. To conclude that teenagers with ADHD are worse drivers than those without ADHD, data collected in the baseline condition (no distraction) would need to show that those with ADHD performed significantly worse than those without ADHD, and this is not suggested by the passage.

Educational objective:

Attention-deficit hyperactivity disorder (ADHD) is characterized by inattention (distractibility, difficulty focusing) and impulsivity (acting without thinking about consequences).

Time spent:0 sec

QID:400758

Subject:

System:

Learning, Memory, and Cognition

Behavioral Sciences

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Which type of memory is LEAST likely to be engaged during the driving simulation?

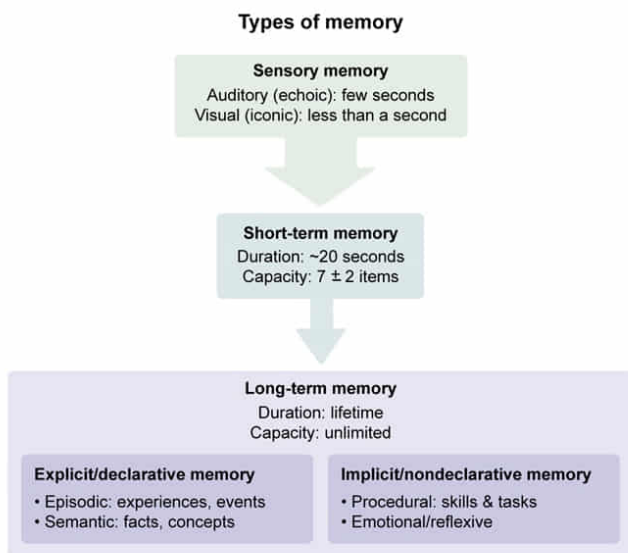
☐ A. Implicit [7%]

- ☐ B. Sensory [3%]
☐ C. Short-term [10%]
✔ ☒ D. Episodic [78%]

Correct

78% answered correctly

Explanation:



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There are three main types of memory, with varying durations and capacities: sensory, short-term, and long-term.

Sensory memory has the shortest duration and smallest capacity; it is a very temporary store for perceived sensations. A majority of sensory information is not transmitted to short-term memory and is lost immediately.

Short-term memory has a short duration and a storage capacity of about seven items (plus or minus two). Maintenance rehearsal (mentally repeating something over and over) can prolong the duration of short-term memory. Information not transferred to long-term memory is lost.

Long-term memory is thought to be a permanent store with an infinite storage capacity consisting of two branches: explicit memory and implicit memory. Explicit memories can be consciously recollected and put into words (eg, factual information, personal events), while implicit memories cannot (eg, smells).

There are two types of explicit memory. **Semantic** memory consists of facts and general knowledge (eg, presidents' names), whereas **episodic** memory consists of memories of personal experiences (eg, what you wore your first day of college). Episodic memory pertains *least* to the study described because the study did not focus on the recollection of personal experiences.

(Choice A) Implicit memory is memory stored without effort or awareness. The main type of implicit memory is procedural memory, or memory of how to do things (eg, ride a bicycle, tie a shoe). Driving relies on implicit memory and was thus engaged during this study.

(Choice B) Processing perceptions (sights, sounds) while driving relies on sensory memory and was thus engaged during this study.

(Choice C) Driving relies on short-term memory, including any information that must be retained longer than the sensory memory duration allows (eg, position of cars in others lanes), and was thus engaged during this study.

Educational objective:

Sensory memory has the shortest duration and smallest capacity, short-term memory lasts for half a minute, and long-term memory is permanent. The two types of long-term memory are explicit (declarative) and implicit (procedural). The two types of declarative memory are episodic (personal experiences) and semantic (facts).

Time spent:0 sec
Subject:
Behavioral Sciences

QID:400759
System:
Learning, Memory, and Cognition

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Which aspect of working memory appears to be overloaded, according to the results of this study?

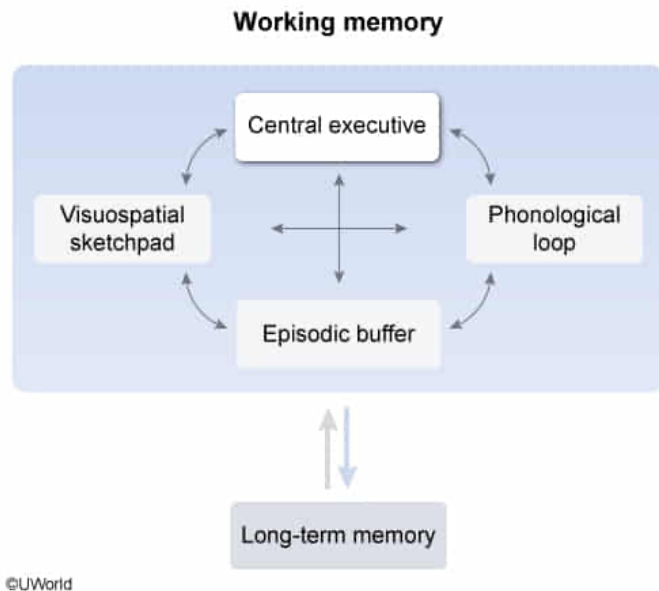
☐ A. Phonological loop [1%]

- ☐ B. Visuospatial sketchpad [51%]
- ☐ C. Episodic buffer [2%]
- ✓ ☒ D. Central executive [43%]

Correct

44% answered correctly

Explanation:



Working memory and short-term memory are distinct executive functions. While both involve a limited and temporary store, **working memory** involves the **manipulation** and **processing** of information while short-term memory does not.

Working memory is conceptualized as consisting of a **central executive**, which regulates attention and task switching, and three subsystems, which are controlled by the central executive. The **visuospatial sketchpad** is employed when manipulating visual and/or spatial information (eg, reading a map). The **phonological loop** is employed when manipulating spoken and written information (eg, reading a book). The **episodic buffer** is responsible for temporal processing (understanding the timeline of events) and integrating information from long-term memory into working memory (eg, remembering how to multiply when figuring out a tip at a restaurant).

The results indicated that driving was impaired because of the cognitive overload produced by competing tasks and visual inattention. The central executive, responsible for attention and switching among various tasks,

such as holding a conversation and driving, appears to be the aspect of working memory that is overloaded.

(Choices A and B) Processing linguistic information during both distraction conditions (auditory and texting) would involve the phonological loop. Processing spatial information required for driving tasks would involve the visuospatial sketchpad. However, the central executive is responsible for *switching between* linguistic and spatial information processing systems, so overloading of the central executive best accounts for the results of this study.

(Choice C) Integrating information (such as past driving experiences) into working memory to be used in relationship to one's current task involves the episodic buffer; however, this was not directly assessed by the study.

Educational objective:

Working memory consists of the central executive (attention, task switching) and three lower subsystems: the visuospatial sketchpad (visual/spatial information), the phonological loop (linguistic information), and the episodic buffer (timeline, long-term memory integration). The central executive controls the lower subsystems.

Time spent:0 sec
Subject:
Behavioral Sciences

QID:400760
System:
Learning, Memory, and Cognition

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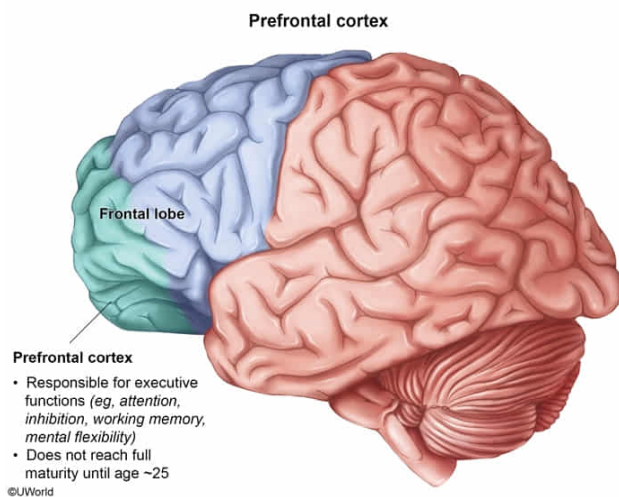
A proponent of the biological perspective would suggest that distracted driving should decrease with age as:

- ✔ ☒ A. the prefrontal cortex develops. [90%]
- ☐ B. individuals are punished for it. [1%]
- ☐ C. it is naturally selected for in the population. [5%]
- ☐ D. individuals become less influenced by peer pressure. [1%]

Correct

91% answered correctly

Explanation:



The **prefrontal cortex**, the anterior portion of the frontal lobe, regulates **executive functions**, including working memory, reasoning, planning, impulse inhibition, attention, making decisions, and solving problems. The prefrontal cortex is still developing during the teens and does not reach full maturity until the mid-twenties.

In psychology, the **biological perspective** seeks to explain human behavior through understanding physical and physiological causes. This perspective suggests that human behavior and mental illness can be explained by examining the structure and function of the brain. For example, a proponent of this perspective would suggest that the impulsive behavior of teenagers is the result of an immature prefrontal cortex.

(Choice B) Behaviorism, which emphasizes the role of punishment and reinforcement on shaping behavior, would suggest that a decrease in distracted driving with age is the result of punishment.

(Choice C) A proponent of the biological perspective might suggest that distracted driving by teenagers, which can result in death, would be naturally selected *against* (not for) in the population.

(Choice D) A proponent of the social cognitive perspective, which emphasizes the role of social influence on behavior, might explain behavior in terms of peer pressure.

Educational objective:

The biological perspective explains behavior in terms of the development, structure, and function of the brain and central nervous system. The prefrontal cortex, which controls executive functions (eg, working memory, reasoning, impulse inhibition, attention), does not reach full maturity until the mid-twenties.

Time spent:0 sec
Subject:
Behavioral Sciences

QID:400761
System:
Learning, Memory, and Cognition

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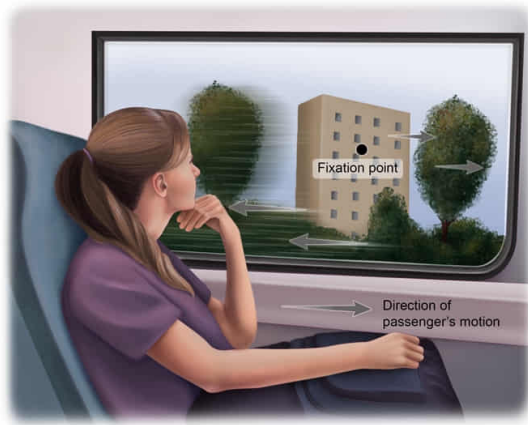
Which of the following processes assists in the perception of depth and motion required during the driving simulation?

- ✓ ☒ A. Motion parallax [41%]
☐ B. Retinal disparity [39%]
☐ C. Convergence [18%]
☐ D. Phi phenomenon

Correct

40% answered correctly

Explanation:



People use a variety of monocular (one eye) and binocular (both eyes) visual cues to perceive depth and motion. Since the driving simulator projected computer-generated two-dimensional visuals on a screen a few feet away from the participants, monocular cues would generate a sense of motion and depth. In other words, since there is no *real* depth or distance, participants are not relying on binocular cues.

Motion parallax (or **relative motion**) is a monocular cue whereby objects in the **foreground** are perceived as **moving faster** than objects in the **background**. Motion parallax is a perceptual process that would not require three-dimensional depth but would still allow subjects to perceive both depth and motion.

(Choices B and C) **Retinal disparity** and convergence are both binocular cues for depth/distance. Retinal disparity occurs because each eye transmits a slightly different image to the brain, which infers distance from the disparity. Convergence is the extent to which the eyes turn inward (converge) to focus on an object; closer objects require more convergence, which helps the brain infer distance. The two-dimensional simulation made the use of binocular cues irrelevant, and neither convergence nor retinal disparity is involved with *motion* perception.

(Choice D) The phi phenomenon (also known as the motion picture effect) is an optical illusion in which a series of still photographs presented in rapid succession appear to be moving. The phi phenomenon may have been relevant to perceiving motion during the simulation but is irrelevant to *depth* perception.

Educational objective:

Monocular (one eye) and binocular (both eyes) cues are important for visual perception. Motion parallax (relative motion) is a monocular cue important for the perception of motion and distance. Motion parallax describes when objects in the foreground appear to be moving faster than objects in the background.

Time spent: 0 sec
Subject:
 Behavioral Sciences

QID: 400762
System:
 Learning, Memory, and Cognition

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Researchers designed the two distraction conditions to require multitasking and predicted that the hands-free conversation condition would be less distracting than the texting condition because of:

- ✔ ☒ A. task dissimilarity. [63%]
- ☐ B. the cocktail party effect. [7%]
- ☐ C. the interference effect. [25%]
- ☐ D. speech shadowing. [3%]

Explanation:

Divided attention (multitasking)		
Task similarity	Easy: dissimilar tasks (eg, listening to audiobook while running)	Hard: similar tasks (eg, listening to audiobook while writing a paper)
Task difficulty	Easy: simple tasks (eg, tapping foot while clapping hands)	Hard: challenging tasks (eg, performing surgery while doing complex mental math)
Task practice	Easy: well-practiced tasks (eg, talking on the phone while walking)	Hard: unfamiliar tasks (eg, practicing a new language while learning to knit)

Attention, cognitive processes that focus on certain stimuli while filtering out others, can be divided or selective. **Divided attention** ("multitasking") describes attending to **more than one** stimulus/task at a time, while selective attention describes focusing on one stimulus and ignoring others.

Multitasking is impacted by **task similarity**, **difficulty**, and **practice**. It is easier to perform two dissimilar tasks than two similar tasks. Driving, which requires visual attention, is easier to do while engaging in a hands-free call (auditory attention) than while texting, because both texting and driving require visual attention.

Tasks that are easier to perform are also easier to do simultaneously. When driving a familiar route in nice weather, it is easy for the driver to simultaneously hold a conversation. However, if driving an unfamiliar route in terrible weather, the driver may ask passengers to stop talking.

Tasks that are well practiced are also easier to do simultaneously. A 45-year-old may have no problem driving and talking on the phone, whereas a 16-year-old who is still learning to drive will find it more difficult to do both.

(Choice B) The cocktail party effect describes when attention quickly shifts from an attended stimulus to an unattended stimulus when something significant occurs. At a crowded party you tune out all competing noise to focus on the person you are talking to (attended stimulus), but if you hear your name in another conversation, your attention quickly shifts to that unattended stimulus. This is a

selective attention (not multitasking) process.

(Choice C) Interference is a memory (not attention) process describing when old information prevents recollection of new information (proactive interference) or new information prevents the recollection of old information (retroactive interference).

(Choice D) Speech shadowing is a selective attention (not multitasking) process used in dichotic listening tasks (competing information presented in each ear) that involves repeating information presented in one ear while tuning out the competing information in the other ear.

Educational objective:

Divided attention (multitasking) occurs when an individual attends to more than one stimulus or task simultaneously. The ability to multitask is significantly impacted by task similarity, difficulty, and practice. Tasks that are dissimilar, less difficult, and/or well practiced are easier to perform simultaneously.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400763

System:
Learning, Memory, and Cognition